

Source-Grounded Research

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Abstract

The most expensive mistake an AI system can make is not being wrong. It is being *confidently wrong in a way you cannot trace*. A model that fabricates a statistic costs you a correction. A model that fabricates a statistic, attributes it to a regulator, and files it under "verified" costs you a decision — and possibly a reputation.

This paper is about the discipline that prevents that: **source-grounded research**. It is the practice of tying every claim to where it came from, distinguishing what you observed directly from what someone told you, and stating plainly how sure you are. These are not academic niceties. They are the load-bearing beams of any knowledge system you would trust to inform real decisions — a due-diligence memo, a regulatory read, or the searchable public-work knowledge base that is your first practical wedge.

We cover five concepts: **primary source**, **secondary source**, **citation**, **provenance**, and **uncertainty**. We ground each in three running examples — a Galoy product brief, a regulatory signal brief, and public-company research — and then connect them to how a system like FRIDAY should behave. The claim of the episode is narrow and defensible: *research quality is not a function of how much a system knows. It is a function of how honestly it tracks where its knowledge came from and how much to trust it.*

1. Why this matters before you build anything

You are building an operating layer that will eventually make or shape decisions. The decisions will only be as good as the research underneath them, and the research will only be as good as its weakest habit. If the habit is "generate fluent text that sounds authoritative," you have built a machine for producing plausible errors at scale. That is worse than no machine, because plausible errors are the hard ones to catch.

Consider the difference between two answers to the same question — "*What is Galoy's core product?*"

Answer A: "Galoy is a leading Bitcoin banking platform trusted by financial institutions worldwide to deliver next-generation digital finance."

Answer B: "Galoy builds open-source banking infrastructure for Bitcoin and the Lightning Network. Its flagship deployment is Bitcoin Beach Wallet in El Zonte, El Salvador [galoy.io/about, retrieved 2026-06]. I could not confirm the number of institutional deployments; the website lists case studies but no count [uncertain]."

Answer A is marketing. It has no sources, no provenance, and — tellingly — no admission of what it doesn't know. Answer B is research. It is less impressive to read and far more useful to act on. The whole of this episode is the difference between A and B, made systematic.

The move you are learning is unglamorous on purpose. Serious people in serious rooms do not reward confidence. They reward the ability to say "*here is what I know, here is how I know it, and here is where the ground gets soft.*" A system that can do that earns the right to be in the room.

2. Primary versus secondary: the first cut

Every piece of evidence sits somewhere on a chain that runs from an event to your desk. **The closer to the origin, the more you can trust it — and the fewer hands it has passed through to distort it.**

A **primary source** is the thing itself, or the closest available record of it. The company's own filing. The regulator's published rule. The founder's own words in a talk. The raw dataset. When Galoy publishes its architecture on its own documentation site, that is primary: it is the organization speaking directly about itself.

A **secondary source** describes, summarizes, or interprets a primary one. A journalist's article about Galoy's funding round. An analyst note explaining a new regulation. A blog post summarizing an earnings call. Secondary sources are not inferior — they are often where the *interpretation* lives, and interpretation is valuable. But they carry the fingerprints of whoever made them. A summary can compress, and compression loses information. A summary can also be wrong, and its wrongness inherits your trust if you don't check.

There is a quiet trap here worth naming. The distinction is not about *quality* — a careful secondary source can be more reliable than a sloppy primary one. It is about *distance from the origin*. A press release is primary (the company said it) but self-interested. A skeptical investigative article is secondary but may be more truthful. Grounding means tracking both the distance and the direction of the pull.

Applied to the three examples:

- **Galoy product brief.** Primary: Galoy's own repos, docs, and site. Secondary: coverage in Bitcoin trade press, a Lightning ecosystem report. Build the brief from primary sources; use secondary ones to catch what the company won't say about itself.
- **Regulatory signal brief.** Primary: the actual text of the rule, the agency's own notice, the official register entry. Secondary: a law firm's client alert interpreting it. The rule is what binds you. The alert is someone's reading of what binds you — useful, but do not mistake the reading for the rule.
- **Public-company research.** Primary: the 10-K, the earnings call transcript, the SEC filing. Secondary: analyst reports, news, the summary a model gives you. For anything that could move a decision, the number lives in the filing, not the article about the filing.

The rule of thumb: **for any claim that carries weight, find the primary source or flag that you couldn't.** A research system that cannot tell you which of its claims rest on primary evidence is not doing research. It is doing paraphrase with good posture.

3. Citation: the address, not the ornament

A **citation** is a pointer from a claim to its source, specific enough that someone else can go check. That last clause is the entire point. A citation is not a decorative footnote that makes text look scholarly. It is an *address* — a way for a human, or another system, to walk back to the origin and verify.

This distinguishes a real citation from the thing models often produce, which is a plausible-looking reference to something that may not exist. When a language model invents a citation, it is not lying in any meaningful sense — it has no concept of the source existing or not. It is generating text that has the *shape* of a citation because citations tend to appear in that context. This is why "the model cited a source" means nothing on its own. The question is whether the source exists, says what the model claims, and actually supports the specific point.

A citation earns its keep when it satisfies three tests:

1. **It resolves.** The link works, the document exists, the page is real.
2. **It contains the claim.** The cited source actually says the thing, not something adjacent.
3. **It supports the specific point.** Not "this article is generally about the topic" but "this sentence backs this exact assertion."

Most bad citations pass the first test, fail the third, and never get checked. A system that cites the earnings call for a revenue figure that the call never stated has committed the most common and most dangerous research error: a citation that lends borrowed authority to an unsupported claim. The citation makes the error *harder* to catch, because it looks handled.

For your work, this reframes what "good AI research" looks like. It is not the volume of citations. A wall of references is easy to generate and impressive to nobody who checks. It is the *fidelity* of each citation to the claim it anchors. One correct, specific, verifiable citation is worth a page of decorative ones.

4. Provenance: the chain of custody

If citation points to *where a claim came from*, **provenance** is the full record of *how the claim traveled to you* — every transformation along the way. It is a chain-of-custody concept, borrowed from law and forensics, and it is the piece most AI systems get wrong because it is the piece that requires infrastructure, not just good intentions.

Consider a single fact moving through a system: a regulator publishes a rule → a model retrieves a summary of it → that summary gets stored in a knowledge base → weeks later, the system surfaces the summary in a brief. By the time it reaches the brief, four things have happened to that fact, and *any* of them could have introduced error. Provenance is the record that lets you answer: *Where did this originally come from? What touched it? When? Was it the primary text or someone's summary of it? Has the source changed since?*

Provenance is what turns a claim from a floating assertion into an auditable one. Without it, a knowledge base degrades into a pile of confident sentences with no way to tell the fresh from the stale, the primary from the third-hand, the verified from the guessed. With it, you can trace any output back to its roots and re-check the roots when they matter.

This is where the mechanism underneath modern research tools becomes worth understanding — briefly, because it is adjacent to our thread. Tools like NotebookLM and Perplexity work by **retrieval-augmented generation** (RAG): rather than answering from the model's trained-in memory, the system first *retrieves* relevant source passages, then generates an answer grounded in those passages, and links back to them. The retrieval step is what makes citation possible at all — the model is answering *about specific documents it just pulled*, not reciting from a blurry internal average.

How does retrieval find the right passages? Through **semantic similarity** — comparing the *meaning* of your query to the meaning of stored text, measured as **vector distance** in a mathematical space, not keyword overlap and certainly not any physical or geographic closeness. This matters for one honest reason: **similarity is not truth**. The system retrieves passages that are *related* to your question. Related is not the same as correct, and it is not the same as sufficient. A passage can be the closest match in the entire corpus and still be wrong, outdated, or beside the point. Embeddings represent similarity; they do not certify accuracy. Keep that seam visible.

Provenance is the discipline that survives after the retrieval is done. It says: *even if the retrieval was good, record where each piece came from, so that when the source changes or the claim is challenged, you can go back*. This is your wedge, stated precisely — a public-content knowledge base is only *operationally useful* if every reusable

answer carries its provenance. Searchable without provenance is a search engine. Searchable *with* provenance, citation, and freshness is a research instrument.

5. Uncertainty: saying how sure you are

The final concept is the one most systems skip, because it is the least flattering. **Uncertainty** is the honest labeling of how confident a claim is and why. It is the difference between "revenue was \$2.1B" and "revenue was approximately \$2.1B [from the Q3 call; full-year figure not yet filed, so this is a run-rate estimate]."

Uncertainty is not weakness. It is calibration — the alignment between how sure a system *sounds* and how sure it has *earned the right to be*. An uncalibrated system says everything in the same confident register, whether it is quoting a filing verbatim or extrapolating from a half-remembered summary. That flat confidence is precisely what makes it dangerous, because you cannot tell the load-bearing claims from the guesses by reading them. They all sound the same.

A well-built research system marks its uncertainty in at least three forms:

- **Sourced and verified** — traced to a primary source, the source checked. State plainly.
- **Sourced but soft** — from a secondary source, or a primary source that is ambiguous, self-interested, or stale. State with the caveat attached.
- **Inferred or unknown** — reasoned from incomplete evidence, or simply not found. Say so explicitly. "*I could not confirm this*" is a complete and respectable answer.

There is a deeper point here about what language models are and are not. An LLM is a reasoning and generation engine with real limits; it does not *know* things the way a person does, and it has no built-in sense of its own confidence that maps to truth. Its fluency is constant regardless of whether it is right. So uncertainty cannot be left to the model's tone — it must be *engineered in* through the surrounding system: retrieval that reports when it found nothing, checks that flag unsupported claims, and gates that require a human to sign off before a soft claim becomes an action. Fluency is not confidence, and confidence is not correctness. A serious system keeps those three apart on purpose.

The practical payoff is that labeled uncertainty is what makes an AI research output *usable by a competent human*. When the system tells you which claims are firm and which are soft, you know exactly where to spend your scarce attention. You verify the three soft claims and trust the twelve firm ones. When everything is asserted with equal confidence, you either have to re-check all fifteen — in which case the system saved you nothing — or check none, in which case it will eventually burn you.

6. The five concepts, in one view

Concept	Plain-language definition	The question it answers	Failure mode when absent
Primary source	The origin: the thing itself or its closest record	<i>How close to the event is this?</i>	Building on paraphrase; inheriting others' distortions
Secondary source	A description or interpretation of a primary source	<i>Whose reading am I trusting?</i>	Mistaking a summary for the original; missing the spin
Citation	A specific, checkable pointer to a source	<i>Can someone verify this exact claim?</i>	Fabricated or decorative references that lend false authority

Concept	Plain-language definition	The question it answers	Failure mode when absent
Provenance	The full chain of how a claim traveled to you	<i>What touched this, and when?</i>	Stale, untraceable knowledge; no way to re-check
Uncertainty	Honest labeling of confidence and its basis	<i>How sure am I, and why?</i>	Flat confidence; guesses indistinguishable from facts

7. Where this lives in FRIDAY

Now the connection, made only after the mechanism is clear. FRIDAY is meant to be an orchestrator and operator, not a chatbot that emits fluent paragraphs on request. GBrain is the knowledge and memory substrate underneath it. The five concepts in this paper are the quality standard for what GBrain is allowed to hold and what FRIDAY is allowed to say.

Concretely, a research-grade knowledge substrate should enforce a few rules:

- **Every stored claim carries its source and its retrieval date.** No orphan facts. If GBrain cannot say where a claim came from, GBrain should not present it as known.
- **Primary and secondary are tagged, not blended.** When FRIDAY assembles a Galoy brief, it should be able to show you which lines rest on Galoy's own documents and which rest on third-party coverage.
- **Citations are verified, not just generated.** A retrieval step that links to real passages, plus a check that the passage actually supports the claim — not a model asserting that it does.
- **Uncertainty is a field, not a footnote.** Confidence travels with the claim through the whole pipeline, so a soft fact stays visibly soft even three steps downstream.
- **Human approval gates guard the soft claims.** Automation without governance is not efficiency; it is unsupervised error. A brief built from soft or inferred claims should require a human sign-off before it becomes a decision. The point of governance is not to slow you down — it is to keep the automation trustworthy enough to be worth having.

None of this makes FRIDAY a "magic employee." It makes it a disciplined research assistant whose work you can audit — which is a far more valuable and far more buildable thing. The wedge you have chosen (making a person's or company's public body of work searchable, reusable, cited, and operationally useful) is this discipline applied to a bounded corpus. Get the five concepts right on one public knowledge base and you have the template for the rest.

8. The main failure mode

If you remember one risk, remember this one: **the ungrounded claim that survives because it looks handled.**

It works like this. A model produces a fluent, well-structured answer with citations attached. The citations look real. The prose is confident and calibrated-sounding. Everything about the *form* signals "this was researched carefully." But one citation points to a source that doesn't say what's claimed, and the claim itself is the one that matters. Because the output looks handled, nobody checks. The error propagates into a brief, the brief into a decision.

This failure is dangerous precisely *because* of the machinery we've praised. Retrieval, citation, and fluent generation all make outputs *look* more trustworthy. If the verification underneath is weak, you have built a system that is better at *appearing* grounded than at *being* grounded — which is strictly worse than an obviously sloppy system, because the

sloppy one at least warns you.

The defense is not more sophistication. It is the boring practice of this entire paper: check whether the primary source exists and says the thing, keep provenance so you can re-check, and label uncertainty so the soft claims announce themselves. A system that does these three things will *look* less impressive than one that emits confident walls of cited text. It will also be the one you can actually stake a decision on.

9. Vocabulary

- **Primary source** — The origin of a claim: the thing itself or the closest available record of it (a filing, a rule's text, raw data, the founder's own words).
- **Secondary source** — A description, summary, or interpretation of a primary source. Useful for interpretation; carries the maker's fingerprints.
- **Citation** — A specific, checkable pointer from a claim to its source. Real only if it resolves, contains the claim, and supports the exact point.
- **Provenance** — The full chain of custody: where a claim originated, what transformed it, and when. What makes a knowledge base auditable rather than a pile of confident sentences.
- **Uncertainty** — Honest labeling of how confident a claim is and why. Calibration between how sure a system sounds and how sure it has earned the right to be.
- **Retrieval-augmented generation (RAG)** — Answering by first retrieving relevant source passages, then generating a response grounded in and linked to them, rather than from the model's internal memory alone.
- **Semantic similarity / vector distance** — Comparing meaning as position in a mathematical space. How retrieval finds relevant passages. *Similarity is not truth.*
- **Calibration** — The alignment between expressed confidence and actual reliability. The goal of good uncertainty labeling.

10. Reading guide

Two anchors, read with a specific question in mind rather than end to end.

NotebookLM overview — <https://notebooklm.google/>

The useful thing to notice here is *grounding to a fixed source set*. NotebookLM answers about documents you give it and cites back into them, rather than answering from the open internet or trained-in memory. As you read, ask: *what does it do when the answer isn't in the sources?* That behavior — refusing to fabricate outside its corpus — is the primary-source discipline of Section 2 turned into a product constraint. This is the closest public analog to what a well-behaved GBrain query should feel like. **~15 minutes.**

Perplexity Pages / examples — <https://www.perplexity.ai/>

Here the thing to study is *citation as a first-class output*. Perplexity attaches inline references and lets you inspect them. Run one query on something you already know cold, then *actually click through* two or three citations and check them against Section 3's three tests: does it resolve, contain the claim, and support the specific point? You will likely find one that passes the first test and fails the third. That gap — between a citation that exists and a citation that supports its claim — is the entire lesson of this episode, sitting in a live product. **~15 minutes, most of it clicking through, not reading.**

Do not read anything else for this. The point is not coverage; it is to see the two concepts operating in tools you can touch.

11. Walking questions

1. Which of these five concepts could you explain clearly to a skeptical investor right now, with an example — and which would you fumble?
2. Where is the line between primary and secondary still blurry for you? (Is a founder's blog post primary or secondary? A press release? An analyst's model?)
3. In FRIDAY or GBrain, where does an ungrounded claim currently have the easiest path to surviving unchecked?
4. What is one bounded, low-stakes workflow — a single Galoy brief, one regulatory signal, one public-company one-pager — where you could test full grounding (primary sourcing, real citations, provenance, labeled uncertainty) end to end, and see what breaks?

One-sentence takeaway

Research quality is not how much a system knows — it is how honestly it tracks where each claim came from and how sure it has the right to be.